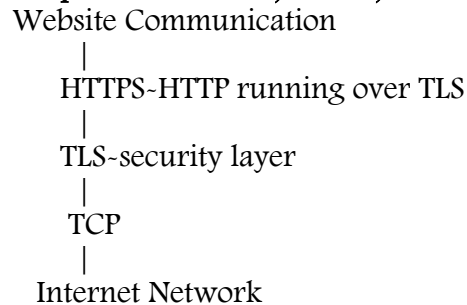


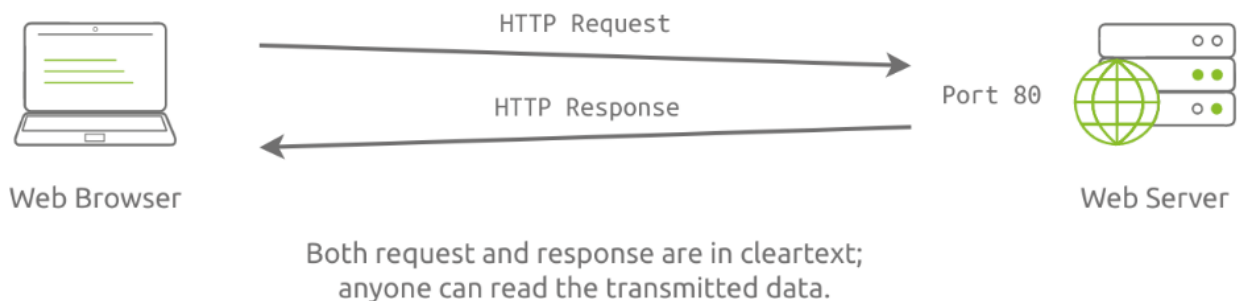
Networking Secure Protocols

- **TLS**-(Transport layer Security) is a technology that keeps information private and safe when it travels across the internet.
TLS- Encrypt, authenticate and integrity.
Its encounter are:secure websites HTTPS,online banking,email services,messaging apps, Etc
- **HTTP**-Hypertext Transfer Protocol Secure-its protocol browsers uses to communicate with the websites.with https passwords are encrypted information protected and data cannot be altered.

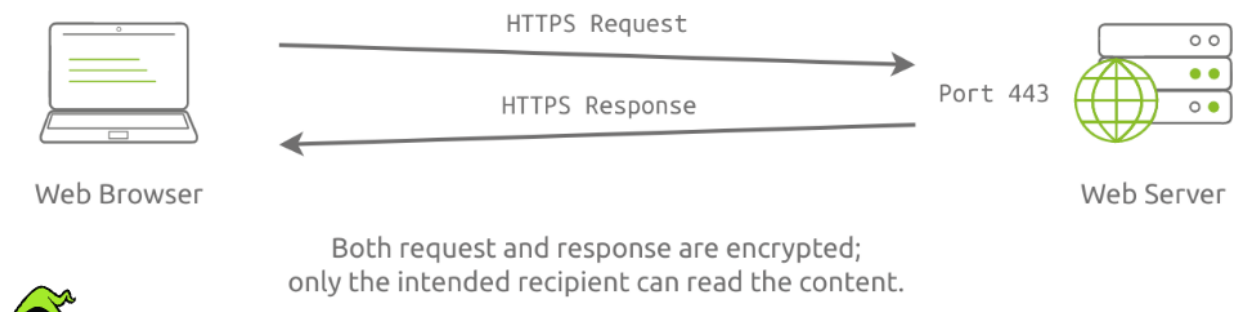
Relationship between HTTP, HTTPS, and TLS



Request and Response - HTTP



Request and Response - HTTPS



- **FTPS Protocol**(File transfer protocol)its protocol which redefines and secure version of [FTP.USED](#) TO transfer files between computers over internet

How FTPS works

1. A client connects to an FTP server.
2. TLS encryption is established.
3. The client logs in securely.
4. Files are uploaded or downloaded through the encrypted connection.

- **SMTP, POP3, and IMAP~**

- **SMTPS**: Secure Mail Transfer Protocol over TLS.
- **POP3S**: Secure Post Office Protocol (email retrieval) over TLS.
- **IMAPS**: Secure Internet Message Access Protocol (email synchronization) over TLS.

The insecure versions use the default TCP port numbers shown in the table below:

Protocol	Default Port Number
HTTP	80
SMTP	25
POP3	110
IMAP	143

The secure versions, i.e., over TLS, use the following TCP port numbers by default:

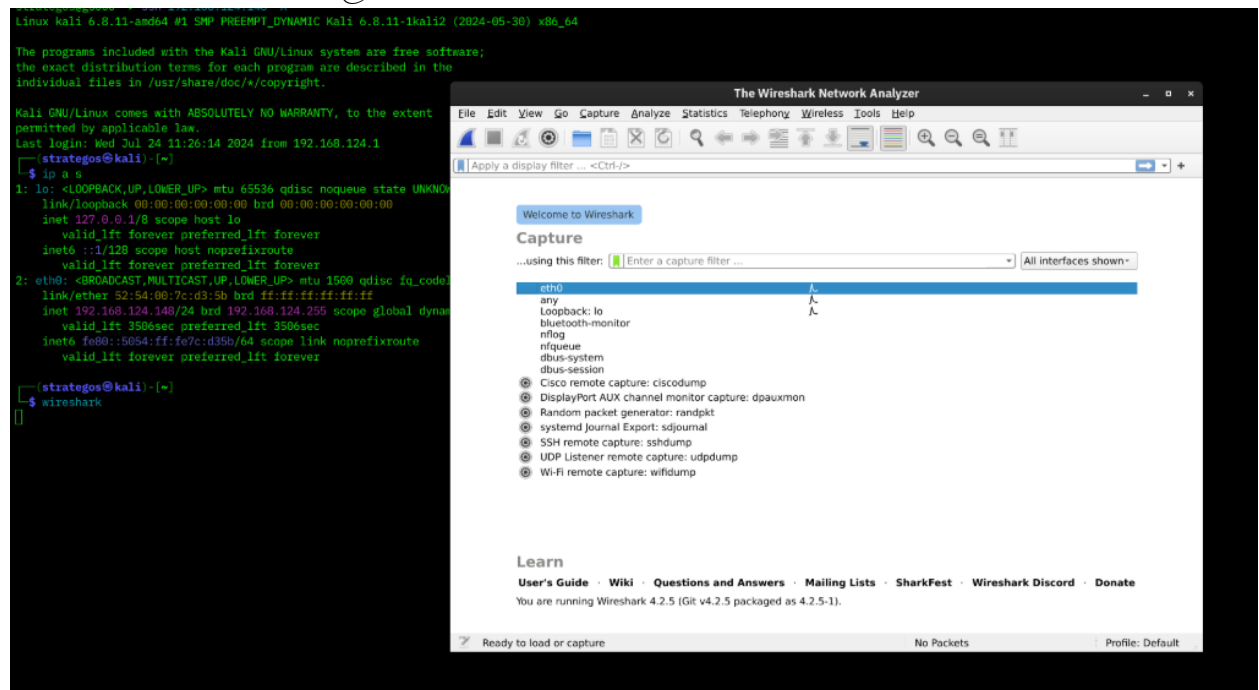
Protocol	Default Port Number
HTTPS	443
SMTPS	465 and 587
POP3S	995
IMAPS	993

- SSH-(Secure shell) protocol used to securely log in to remote systems.

OpenSSH offers several benefits. We will list a few key points:

- **Secure authentication:** Besides password-based authentication, SSH supports public key and two-factor authentication.
- **Confidentiality:** OpenSSH provides end-to-end encryption, protecting against eavesdropping. Furthermore, it notifies you of new server keys to protect against man-in-the-middle attacks.
- **Integrity:** In addition to protecting the confidentiality of the exchanged data, cryptography also protects the integrity of the traffic.
- **Tunneling:** SSH can create a secure “tunnel” to route other protocols through SSH. This setup leads to a VPN-like connection.
- **X11 Forwarding:** If you connect to a Unix-like system with a graphical user interface, SSH allows you to use the graphical application over the network.

Command `ssh user name@hostname` to connect to SSH Server



TELNET server listens on port 23,SSH server listens on port 22.

- **SFTP and FTPS**

SFTP (SSH File Transfer Protocol) and **FTPS** (File Transfer Protocol Secure) both provide secure ways to transfer files, but they use different underlying technologies.

- **SFTP**: Part of the SSH protocol suite, operating on **port 22**.
- **FTPS**: Uses TLS for security, similar to HTTPS, and typically operates on **port 990**.

Example:

```
sftp user@hostname
```

This command establishes a secure file transfer session using SFTP, ensuring that all file uploads and downloads are encrypted.

```
FTP 21 > TTPS 990  
Telnet 23 > SSH 22  
SMTP 25 > SMTPS 465 and 587  
POP3 110 > POP3S 995  
IMAP 143 > IMAPS 993
```